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Paper 1: Association of Neurodevelopmental Disorders and Congenital Anomalies With Prenat

Bibliographic Information

- **Title**: Association of Neurodevelopmental Disorders and Congenital Anomalies With Prenatal Multiple Sclerosis Treatment—Real-World Historical Cohort Study
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- **Journal**: Clinical Pharmacology & Therapeutics, 2025
- **DOI**: N/A

Paper Type: GENERIC

Quick Take

This retrospective cohort study of 1374 children born to mothers with MS found no significant association between prenatal exposure to DMTs (interferon-β, glatiramer, monoclonal antibodies, fumarates, S1PR modulators) and neurodevelopmental disorders. However, exposure to S1PR modulators was associated with a higher rate of major congenital anomalies, though the small sample size limits robustness. The findings provide reassurance regarding neurodevelopmental risks but highlight the need for caution with S1PR modulators.

Executive Summary

In a real-world historical cohort from Clalit Health Services in Israel (2005–2024), 1374 children born to mothers with MS were followed for a mean of 7.5 years. Prenatal exposure to any DMT was not associated with an increased risk of neurodevelopmental disorders (autism spectrum disorder, etc.) in either univariate or multivariate models. Among 1252 children with at least 2 years of follow-up, major congenital anomalies occurred in 3.99% overall. Exposure to S1PR modulators showed a non-significant relative risk of 2.0 (95% CI 0.51–8.1) in the main analysis, but a sensitivity analysis restricted to pregnancy-only exposure yielded a significant RR of 5.16 (95% CI 1.47–18.2). No other DMT class was associated with congenital anomalies. The study is limited by small exposed groups, especially for S1PR modulators, and its retrospective, single-country design.

Scientific Context & Motivation

Multiple sclerosis commonly affects women of childbearing age, yet data on the safety of disease-modifying therapies (DMTs) during pregnancy are sparse. While some DMTs are contraindicated (e.g., S1PR modulators), unplanned pregnancies occur. Prior studies have focused on congenital anomalies, but long-term neurodevelopmental outcomes (e.g., autism) have been largely unexamined. This study addresses both gaps using a population-based Israeli cohort with up to 19.8 years of follow-up.

⚡ Methodological Snapshot

This was a retrospective historical cohort study using the Clalit Health Services database in Israel. The cohort included all children born between 2005 and 2023 to mothers with a diagnosis of MS for at least one year before conception. Exposure was defined as at least one dispensed DMT prescription during the period from 6 months before conception to birth. Seven exposure groups were defined: no exposure, interferon-β, glatiramer, monoclonal antibodies, fumarates, S1PR modulators, and combination therapies. The primary outcomes were neurodevelopmental disorders (including autism spectrum disorder) and major congenital anomalies (as defined by EUROCAT). For neurodevelopmental disorders, a marginal Cox model with robust sandwich estimates was used to account for repeated pregnancies within mothers, adjusting for confounders with $P < 0.1$ in baseline comparisons. For congenital anomalies, a log-binomial regression model estimated relative risks, with adjustment for maternal age and year of birth. Sensitivity analyses restricted exposure to the pregnancy period only (conception to birth) and also matched on birth year.

📋 Detailed Analysis

The study's main strength is its population-based design with long follow-up, enabling assessment of neurodevelopmental outcomes that are rarely studied in this context. The null findings for neurodevelopmental disorders across all DMT classes are reassuring, especially given the known risks with other drugs (e.g., antiepileptics). However, the small number of exposed children (especially for newer agents like S1PR modulators, n=25) limits statistical power. The MCA analysis illustrates this: the main analysis showed a non-significant RR of 2.0, but the sensitivity analysis (restricting to pregnancy-only exposure) yielded a significant RR of 5.16 with very wide confidence intervals. This discrepancy may reflect the inclusion of washout-period exposures in the main analysis diluting the effect, or it could be a chance finding due to small numbers. The authors appropriately note that the two MCAs in the S1PR group were atrial septal defects, which are common in the general population and also appeared in other groups. The biological plausibility is supported by animal studies showing S1P signaling critical for cardiac development. The study's reliance on prescription dispensing rather than actual intake is a limitation, but the low copayment in Israel reduces the likelihood of outside purchasing. The adjustment for OBGYN visits attempts to address surveillance bias, but residual confounding by disease severity (not measured) could influence both treatment choice and outcomes. Overall, the study provides useful real-world evidence but underscores the need for larger, prospective registries.

Domain Context

Multiple sclerosis is a chronic autoimmune disease of the CNS that predominantly affects women during their reproductive years. Disease-modifying therapies (DMTs) reduce relapse rates but have uncertain safety profiles in pregnancy due to exclusion from clinical trials. Current guidelines recommend discontinuing most DMTs before conception, but S1PR modulators (fingolimod, siponimod) are contraindicated due to teratogenicity in animal models. However, real-world data on human outcomes are limited. This study adds to the sparse literature by providing long-term neurodevelopmental follow-up, which is a novel contribution. The findings align with the German MS and Pregnancy Registry showing no increased MCA risk for most DMTs, but the S1PR signal is consistent with the FDA's pregnancy category D designation. The study also highlights the challenge of studying rare outcomes in small exposed populations, a common issue in pregnancy pharmacovigilance.

▯ Key Findings

1. No significant association between prenatal DMT exposure (any class) and neurodevelopmental disorders (crude and adjusted HRs ~0.6–1.4, all P > 0.05). 2. Major congenital anomalies (MCAs) occurred in 3.99% of the cohort. 3. S1PR modulator exposure was associated with a higher MCA rate (8.0% vs. 4.02% in unexposed), with RR 2.0 (95% CI 0.51–8.1) in the main analysis and RR 5.16 (95% CI 1.47–18.2) in the pregnancy-only sensitivity analysis. 4. No other DMT class showed a significant MCA risk. 5. The two MCAs in the S1PR group were atrial septal defects, consistent with known S1P signaling roles in cardiac development.

Strengths & Limitations

Strengths

- Population-based cohort covering >50% of Israeli population
- Long follow-up (mean 7.5 years, max 19.8 years) enabling neurodevelopmental outcome assessment
- Comparison of all major DMT classes within a single study
- Adjustment for multiple confounders (maternal age, SES, comorbidities, etc.)
- Use of robust statistical methods (marginal model for repeated pregnancies, log-binomial regression)

Limitations (Acknowledged by Authors)

- Retrospective design based on administrative data
- Single-country (Israel) limits generalizability
- Small number of exposed children, especially for S1PR modulators (n=25)
- Exposure defined by prescription dispensing, not actual ingestion
- Abortions not included as outcomes, potentially underestimating harmful effects
- Residual confounding remains possible

Limitations (Expert Review)

- No adjustment for MS disease severity or relapse activity during pregnancy
- Potential detection bias due to differential prenatal monitoring (though authors adjusted for OBGYN visits)
- Lack of data on maternal adherence to DMTs
- Wide confidence intervals for S1PR modulator risk estimates limit precision
- Neurodevelopmental disorder diagnosis may vary by age and follow-up duration across groups

Generalizability

The cohort is from a single Israeli HMO with a predominantly Jewish population; results may not generalize to other ethnicities or healthcare systems with different prescribing practices and monitoring protocols. However, the MCA rate (3.9%) is similar to that in the German MS and Pregnancy Registry, suggesting some external consistency.

Figures & Tables

- **Figure 1**: Study timeline showing exposure window (6 months preconception to birth) and follow-up periods for neurodevelopmental disorders and congenital anomalies.
  - *Significance*: Clarifies the exposure definition and follow-up schema used in the analysis.
- **Figure 2**: Kaplan–Meier curves of cumulative incidence of neurodevelopmental disorder diagnoses by DMT exposure group.
  - *Significance*: Shows no apparent separation between groups, supporting the null finding for neurodevelopmental outcomes.
- **Figure 3**: Forest plot summarizing previously published risk estimates for congenital anomalies with various DMTs.
  - *Significance*: Contextualizes the current study's findings within the existing literature.
- **Table 1**: Baseline demographic and clinical characteristics of mothers by DMT exposure group.
  - *Significance*: Highlights differences between groups that were adjusted for in multivariate models.
- **Table 2**: Hazard ratios for neurodevelopmental disorders by DMT exposure (main and sensitivity analyses).
  - *Significance*: Primary result table showing no significant associations.
- **Table 3**: Relative risks for major congenital anomalies by DMT exposure (main and sensitivity analyses).
  - *Significance*: Shows the elevated risk for S1PR modulators, especially in the sensitivity analysis.

Supplementary Materials

Supplementary materials include Figure S1 (study flow chart), Table S1 (excluded teratogenic drugs), Table S2 (list of neurodevelopmental disorders), Table S3 (baseline characteristics for the MCA sub-cohort), Table S4 (list of congenital anomalies), and Table S5 (sensitivity analysis restricted to same birth year).

Future Directions

Prospective, multi-center registries with larger sample sizes are needed to confirm the safety of S1PR modulators and to assess neurodevelopmental outcomes beyond early childhood. Inclusion of abortion data and longer follow-up for rare outcomes would strengthen future studies. Mechanistic studies on S1P signaling in human fetal development could clarify the observed cardiac anomalies.

Expert Commentary

This study fills an important gap by providing long-term neurodevelopmental data, which are reassuring for most DMTs. The signal for S1PR modulators aligns with preclinical evidence and should prompt stricter adherence to contraceptive recommendations. However, the small sample size and wide confidence intervals mean that a clinically meaningful risk cannot be ruled out. Until larger studies are available, the precautionary principle should guide practice.

Bottom Line

For clinicians counseling women with MS who are pregnant or planning pregnancy: Prenatal exposure to DMTs (interferon-β, glatiramer, monoclonal antibodies, fumarates) does not appear to increase the risk of neurodevelopmental disorders. S1PR modulators (e.g., fingolimod) should be avoided during pregnancy due to a potential increased risk of major congenital anomalies, though the evidence is based on small numbers. Shared decision-making should balance the risk of rebound relapse against fetal teratogenicity.

Figures

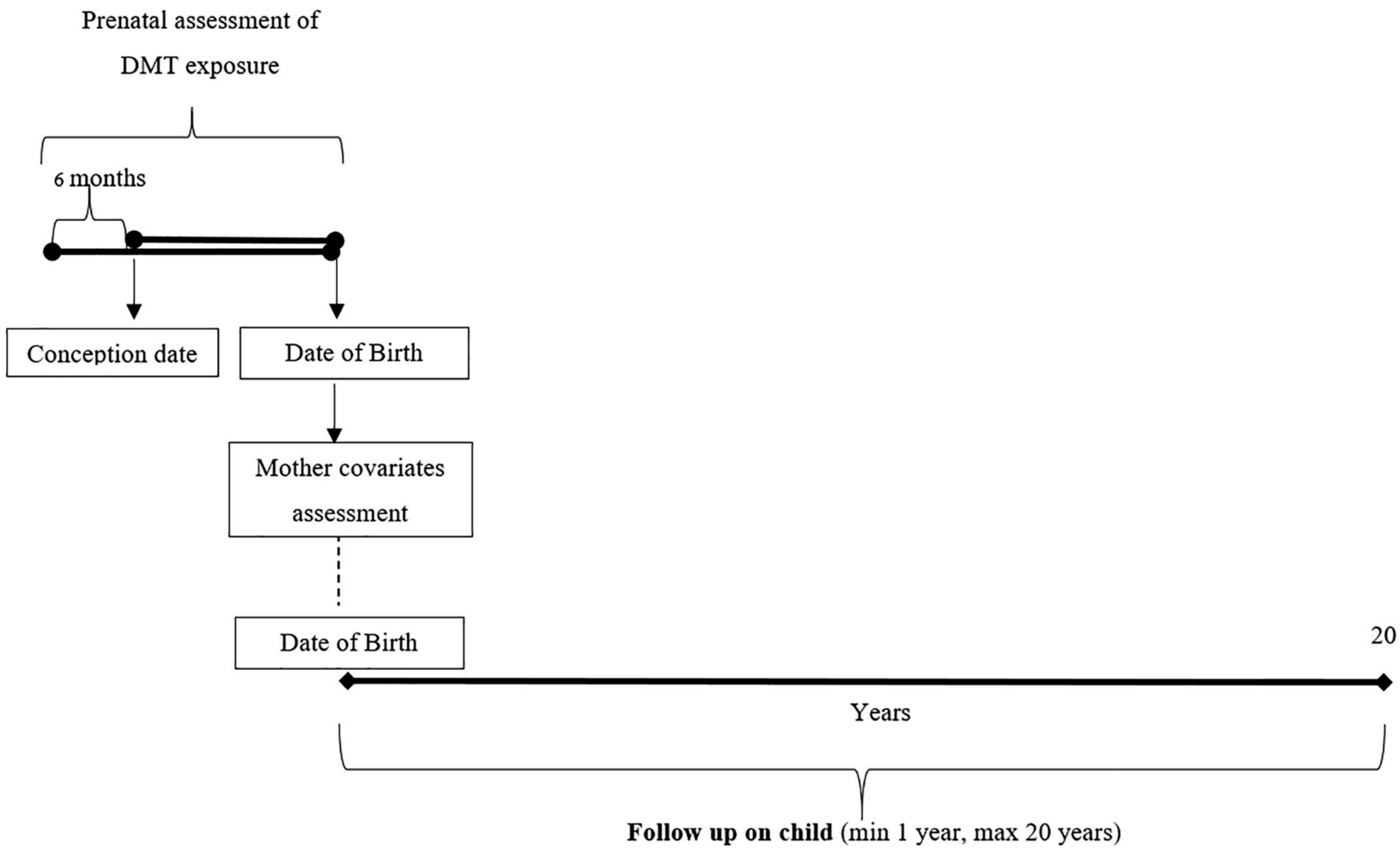

Figure 1: Study timeline.

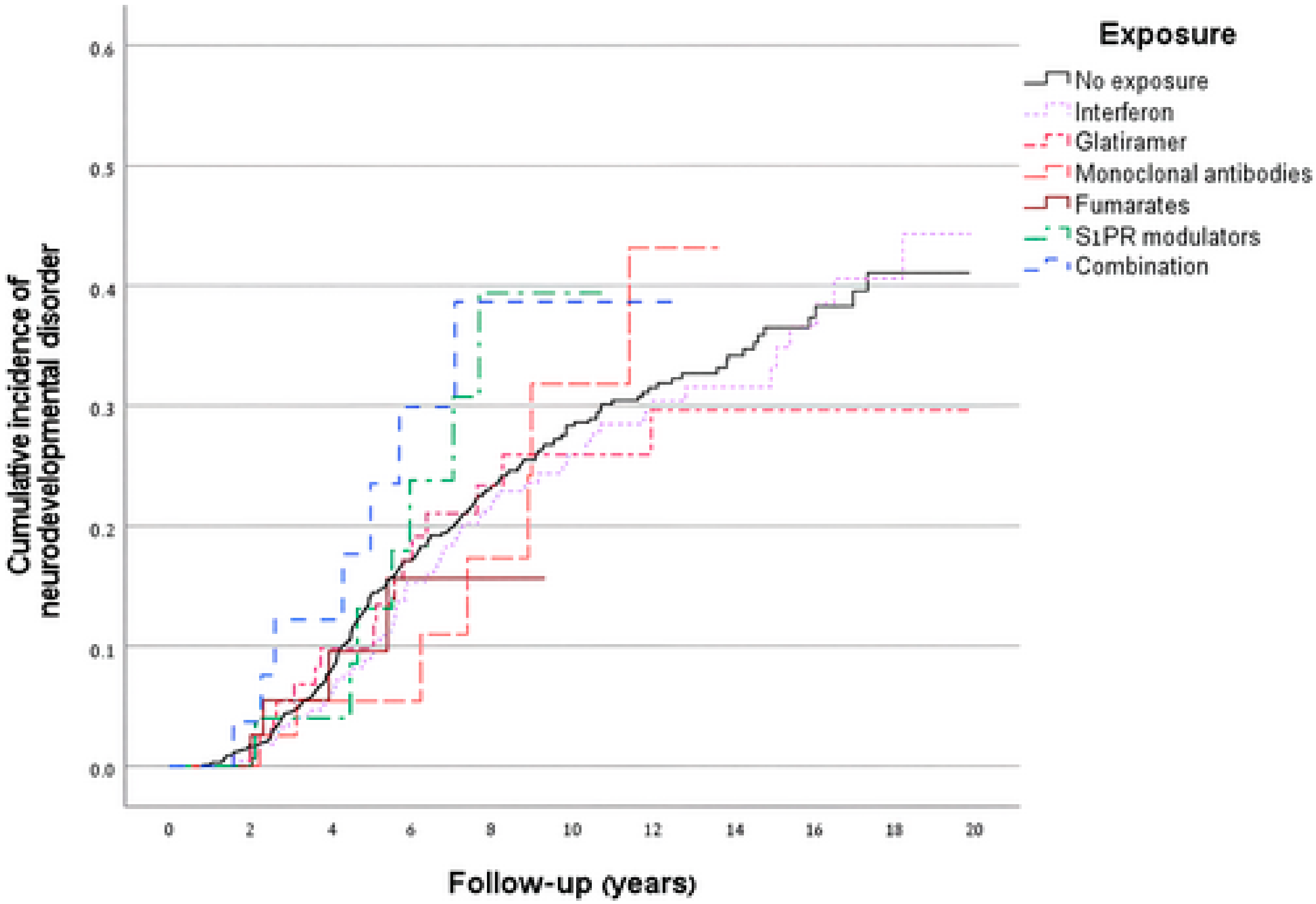

Figure 2: Kaplan–Meier curve. Cumulative incidence of neurodevelopmental disorder diagnoses among children born to mothers with MS, by prenatal exposure to disease-modifyi

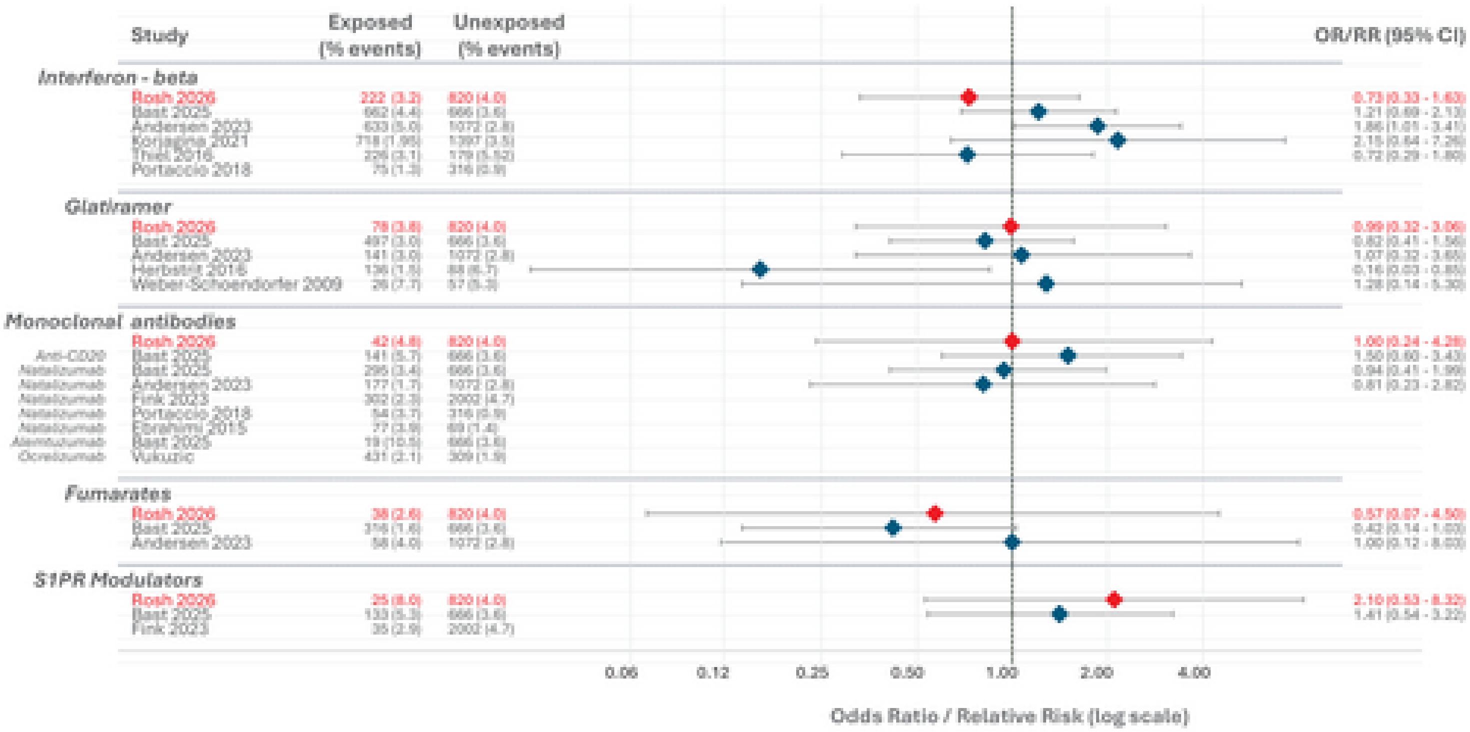

Figure 3: Forest plot showing previously published results from human cohorts describing risks for congenital anomalies, associated with exposure to disese-modifying thera